EMOTION DETECTION FROM A SPEECH SIGNAL USING VIT [VISION TRANSFORMER]

22EC790 Project III

Submitted in partial fulfillment for the requirement of B.E. degree in Electronics and Communication Engineering of Anna University

Submitted by

NISHAANTH R
(Reg. No: 2302917710621091)

HARI HARA SUDHAN C
(Reg. No: 2303917710621032)

PREM KUMAR P
(Reg. No: 2303917710621317)

Guided by

Dr. M.N. Suresh



Department of Electronics and Communication Engineering

THIAGARAJAR COLLEGE OF ENGINEERING
(An Autonomous Institution Affiliated To Anna University, Chennai)

MADURAI – 625 015

NOVEMBER 2025

THIAGARAJAR COLLEGE OF ENGINEERING

(An Autonomous Institution Affiliated to Anna University, Chennai)

MADURAI – 625 015

BONAFIDE CERTIFICATE

This is to certify that the **22EC790 Project III** entitled **"EMOTION DETECTION FROM A SPEECH SIGNAL USING VIT [VISION TRANSFORMER]",** being submitted by **NISHAANTH R (2303917710621091), HARI HARA SUDHAN C (2303917710621032), PREM KUMAR P (2303917710621317)**, in partial fulfillment for the requirement of Bachelor of Engineering Degree in Electronics and Communication Engineering, is a record of Bonafide work done by them during the year 2025-2026 under my supervision. The results embodied in this report have not been submitted to any other university or institute for the award of any degree or diploma.

Dr.M.N.Suresh
(Professor)
Department of ECE,
(Guide)

Professor and Head,
Department of ECE,

Station: Madurai Date:

Certified that the Candidate was examined in the Viva-Voce Examination held at Thiagarajar College of Engineering, Madurai on _____.

INTERNAL EXAMINER EXTERNAL EXAMINER HDECE

LIST OF PAPERS PUBLISHED/PRESENTED

1. Accuracy Enhancement Method For Speech Emotion Recognition From Spectrogram Using Temporal Frequency Correlation And Positional Information Learning Through Knowledge Transfer, "Jeong-Yoon Kim, Seung-Ho Lee",2024.
2. Speech Emotion Recognition from 3D Log-Mel Spectrograms, "Yan Liang, Xincheng Zhang",2021.
3. A Comprehensive Review of SER Systems, "Soukaina El Kharrim, El Mustapha El Hore",2023.
4. Temporal Modelling Matters: A Comparative Study on SER, "Jiaqi Liu, et al.",2024
5. Temporal Modelling Matters: A Comparative Study on SER, "Jiaqi Liu, et al.",2024.
6. MS-SENet: Multi-scale Squeeze-and-Excitation Network for SER, "Mengbo Li, Yuanzhong Zheng",2023.
7. DeepCNN for SER, "Nasir Saleem, Jiechao Gao",2023.
8. Speech Emotion Detection Based on MFCC + CNN-LSTM,Arxiv,2022.
9. Enhancing SER with Deep + Handcrafted Fusion System Design, IJISRT Research Group,2023

ACKNOWLEDGEMENTS

First and foremost, I am grateful to the **GOD ALMIGHTY** for the wisdom, strength, peace, and good health he has bestowed upon us in the completion of this project.

We are extremely grateful to **Dr. L. ASHOK KUMAR,** Principal Incharge, Thiagarajar College of Engineering for sharing the knowledge and technical know-how.

We express my sincere thanks to **Dr. B. MANIMEGALAI,** Professor and Head of Department, Electronics and Communication Engineering for pushing us to new heights and allowing us to prepare the project.

We are highly indebted to our guide **Dr.M.N.SURESH,** Assistant Professor, Electronics and Communication Engineering whose invaluable guidance and motivation given to us throughout the course has shaped the present work as it appears.

If we do not thank our friends for their uplifting inspiration, we will be failing in our duty.

Finally, we extend our heartfelt gratitude to our beloved parents for their unconditional love, sacrifices, and unwavering support throughout our journey. Without them, none of this would have been possible.

Yours faithfully,
Nishaanth R
Hari Hara Sudhan C
Prem Kumar P

LIST OF CONTENTS

Item	Page No.
Title Page	i
Certificates	ii
List of papers published/presented	iii
Acknowledgements	iv
List of contents	v
Abstract	1
1.INTRODUCTION	2
2.LITERATURE REVIEW	6
3.STAKEHOLDER ANALYSIS	13
4. SYSTEM ANALYSIS AND REVERSE ENGINEERING	21
5.PROPOSAL FOR FEATURE ENHANCEMENT	29
6.SUMMARY AND PLAN FOR NEXT PHASE	41
7.IMPROVEMENTS IN PROJECT III	43

ABSTRACT

This project focuses on enhancing the accuracy and efficiency of Speech Emotion Recognition (SER), a Vital task in modern conversational artificial intelligence. Current vision-based SER models, which analyse speech spectrograms, often struggle to capture the complex temporal-frequency relationships crucial for accurate emotional classification, leading to high resource utilization and poor generalization. The proposed methodology addresses these limitations through three core innovations utilizing the Vision Transformer (VIT) architecture.

First, we introduce the use of vertically segmented patches of the log-Mel spectrogram. This forces the VIT to prioritize attending to the correlation of frequency components across the time axis, effectively linking localized vocal features to the exact temporal moment an emotion is expressed. Second, to provide the necessary spatial awareness, we implement Image Coordinate Encoding, a novel absolute positional encoding technique that normalizes and concatenates positional values directly to the image input.

Finally, we employ a Knowledge Transfer (KT) paradigm: a high-resource Teacher Network, complete with a convolutional stem and coordinate encoding, transfers its learned locality and absolute positional knowledge to a streamlined Student Network (a basic VIT) through feature map matching using L1 loss. Experimental results demonstrate that this method not only significantly surpasses state-of-the-art accuracy benchmarks but, crucially, that the lighter Student Network achieves superior efficiency and better addresses the common issue of VIT overfitting. Overall, this work presents a promising, resource-efficient, and high-performance solution for robust SER deployment.

CHAPTER 1:
INTRODUCTION

1.1 BACKGROUND OF THE DOMAIN: Speech Emotion Recognition

1.1.1 Definition and Significance of SER

Speech Emotion Recognition (SER) is a multidisciplinary field focused on analysing and classifying human emotional states from spoken language. It serves as a crucial component in advanced Human-Computer Interaction (HCI) systems, allowing machines to understand the user's affective state, not just the semantic content of their speech. Applications range from intelligent tutoring systems and automated customer service to mental health monitoring and lie detection. Accurate SER is key to creating truly empathetic and adaptive artificial intelligence.

1.1.2 Transition to Vision-Based SER

Traditionally, SER relied on processing hand-crafted acoustic features (e.g., prosody, pitch, energy, Mel-Frequency Cepstral Coefficients - MFCCs). However, recent advancements have favoured a vision-based approach where the raw speech audio is first transformed into a visual representation, such as a log-Mel spectrogram. This transformation allows researchers to leverage highly successful deep learning architectures, particularly Convolutional Neural Networks (CNNs) and Vision Transformers (VITs), which excel at image analysis, for the task of emotion classification. The spectrogram visually encodes frequency and temporal changes, providing a rich, high-dimensional data source.

1.1.3 The Role of the Vision Transformer (VIT) Architecture

The Vision Transformer (VIT), originally developed for natural image processing, has emerged as a powerful tool in vision-based SER. Unlike CNNs, which use local convolution filters, VITs rely on a self-attention mechanism to capture long-range dependencies across the entire spectrogram. The VIT divides the input image (spectrogram) into patches and processes these patches as sequences, making it theoretically excellent at modelling the non-local temporal and frequency relationships that characterize human emotional expression.

1.2 NEED FOR IMPROVEMENT AND STUDY IN VIT-BASED SER

1.2.1 The Challenge of Temporal-Frequency Correlation

While VITs are powerful, when applied to spectrograms, they often treat frequency and time equally, resulting in attention patterns that struggle to focus on the crucial temporal-frequency correlation. Emotional features are often localized: a change in pitch contour (frequency) at a specific moment in time (temporal axis) is essential. Standard square patching and global self-attention dilute this critical information, leading to suboptimal accuracy.

1.2.2 Efficiency and Overfitting in Large VIT Models

VIT models, particularly the large pre-trained variants, require massive amounts of data and computational resources. Deploying these large models in real-world, low-resource environments (e.g., mobile devices, edge computing for call centers) is often impractical. Furthermore, due to the limited size of standard SER datasets (e.g., IEMOCAP, RAVDESS), large VIT models are highly prone to overfitting, learning the specific noise and biases of the training set rather than generalizing well to unseen data. There is a pressing need for a lightweight, yet high-accuracy model.

1.2.3 Lack of Absolute Positional Awareness

The self-attention mechanism is inherently permutation-invariant, meaning it ignores the sequential and spatial order of the input patches. While standard positional encodings are used, they often fail to provide the absolute, coordinate-specific information necessary for effectively distinguishing between temporal location (x-axis) and frequency band (y-axis) location within the spectrogram. This lack of robust absolute positional information hinders the model's ability to localize key emotional cues.

1.3 OBJECTIVES OF PHASE I WORK (FEASIBILITY AND DESIGN)

1.3.1 Primary Objective: Methodological Design for Accuracy Enhancement

The main objective of Phase I is to design and define a novel methodology for VIT-based SER that overcomes the limitations of current systems by enhancing the model's ability to capture temporal-frequency correlations and spatial localization. This includes defining the new patching strategy and the proposed positional encoding technique.

1.3.2 Secondary Objective: Proposing an Efficient Learning Strategy via Knowledge Transfer

A secondary objective is to propose a Knowledge Transfer (KT) framework where a large, feature-rich Teacher Network imparts its refined understanding of locality and positional information to a smaller, more efficient Student Network. The goal is to achieve high accuracy using a lightweight model suitable for practical deployment.

1.3.3 Deliverables of Phase I

The key deliverables for this phase of the project include:

1. A comprehensive Literature Survey establishing the state-of-the-art and identifying clear research gaps.
2. A detailed System Proposal describing the Vertical Patch Segmentation, Image Coordinate Encoding, and the Knowledge Transfer Architecture.
3. A complete Implementation Plan for Semester VI (Phase II) focusing on the development, training, and testing of the proposed model.

CHAPTER 2

LITERATURE SURVEY

2.1 OVERVIEW OF SPEECH EMOTION RECOGNITION (SER) PARADIGMS

2.1.1 Traditional Acoustic Feature Extraction Methods

2.1.1.1 Prosodic Features (Pitch, Energy, and Rate)

Traditional SER heavily relied on prosodic features, which capture the suprasegmental qualities of speech, relating directly to the human vocal mechanism's control over rhythm, stress, and intonation. These features include fundamental frequency (F0 or pitch), energy (intensity/loudness), and speaking rate. Researchers typically extract statistical functionals (e.g., mean, median, maximum, range, and standard deviation) from these raw signals over entire utterances or segments. The appeal of prosodic features lies in their low computational cost and high interpretability, as deviations in pitch contour or intensity directly signal emotional arousal. However, their major limitation is a lack of robustness against variations in speaker, language, recording environment (noise), and the subjective nature of emotional expression, often resulting in performance plateaus around 60-70% accuracy on complex, real-world datasets like IEMOCAP. The simplicity of these features fails to capture the intricate, localized changes in the frequency spectrum that are critical for distinguishing subtle or highly blended emotions, necessitating more complex feature representations.

2.1.1.2 Spectral and Contextual Features (MFCCs and Delta/Delta-Delta)

To improve discriminatory power, researchers shifted focus to spectral features, with Mel-Frequency Cepstral Coefficients (MFCCs) becoming the industry standard for many years. MFCCs model the short-term power spectrum of a sound, specifically approximating the human auditory system's response by using a non-linear Mel scale. An utterance is segmented into short frames (typically 20-30 ms), and a set of 13 to 40 MFCCs are extracted per frame. Crucially, SER performance is significantly enhanced by adding dynamic features, specifically the first and second derivatives of the base coefficients, known as Delta coefficients. These derivatives encode the rate of change and acceleration of the acoustic features over time, providing essential temporal context about how the vocal tract dynamics shift during an emotional expression. While the full feature set (e.g., 39 dimensions: 13 static, 13 Delta, 13 Delta-Delta) provides good performance, it remains limited by the pre-defined, linear statistical relationships the model can learn, often struggling with long-term emotional dependencies and non-linear patterns within the data.

2.1.2 The Evolution to Vision-Based SER

The limitations of traditional, hand-crafted features led to a paradigm shift toward vision-based SER, where the speech signal is pre-processed into a 2D image representation, most commonly the log-Mel spectrogram. The spectrogram plots frequency (y-axis) against time (x-axis) with colour This approach allows the deployment of powerful computer vision models, which can automatically learn complex, hierarchical feature representations directly from the raw spectral data, bypassing the need for explicit feature engineering. intensity representing energy, effectively transforming the SER task into an image classification problem.

2.2 DEEP LEARNING ARCHITECTURES IN VISION-BASED SER

2.2.1 Convolutional Neural Networks (CNNs) and their Limitations

2.2.1.1 CNN Strengths: Locality and Inductive Bias

Convolutional Neural Networks (CNNs) achieved the initial major breakthrough in vision-based SER due to their strong inductive biases for locality and spatial invariance. The convolutional filter naturally focuses on small, local regions of the spectrogram, efficiently identifying features such as harmonic peaks, formants, and energy bursts that are localized in time and frequency. This capability is highly valuable for speech, as many emotional cues are tied to very short, intense events. Furthermore, the weight-sharing property of CNNs drastically reduces the number of trainable parameters, improving efficiency and reducing the data requirements compared to fully connected layers, making them a robust choice for smaller, domain-specific SER datasets. Their hierarchical structure allows for the progressive extraction of low-level acoustic details in early layers and more abstract, high-level emotional patterns in deeper layers, a major advantage over simple acoustic models.

2.2.1.2 CNN Weaknesses: Global Context Modelling

Despite their success, CNNs possess inherent weaknesses when dealing with the full, global context of an emotional utterance, which is crucial for distinguishing emotions that evolve slowly over time. The fundamental operation of convolution is limited to a local receptive field, meaning capturing long-range dependencies (e.g., the relationship between the onset of a sigh and the subsequent word's pitch drop) requires stacking many layers, which leads to increased computational cost and difficulty in training deep networks. This forces the model to stitch together global information indirectly, often resulting in sub-optimal modelling of the entire emotional trajectory across the spectrogram's time axis. This limitation directly impedes the modelling of emotional prosody, which is defined by these large-scale, temporally extended patterns, making the global self-attention mechanism of the Transformer architecture particularly attractive.

2.2.2 The Rise of the Vision Transformer (VIT)

2.2.2.1 VIT Fundamentals: Patching and Self-Attention

The Vision Transformer (VIT) represents a complete departure from the CNN paradigm by replacing convolutions entirely with the self-attention mechanism. The input spectrogram is partitioned into a sequence of fixed-size, non-overlapping patches, which are then linearly embedded and combined with positional encodings. The core of the VIT is the multi-head self-attention (MSA) block, which calculates the relevance (attention weight) between every patch and every other patch across the entire input sequence. This process allows the VIT to model global dependencies directly and in parallel, efficiently linking distant regions of the spectrogram that are acoustically relevant to the expressed emotion. Its strength lies in its ability to discover nuanced, non-local features that CNNs miss, particularly in tasks requiring an understanding of global data structure.

2.2.2.2 Challenges in Applying Standard VIT to Spectrograms

The direct application of standard VIT architectures, which often use square patches, to spectrograms presents two critical challenges. Firstly, the standard square patching fails to account for the physical meaning of the axes: the frequency (y) axis and the time (x) axis. Spectrograms are not isotropic images like natural photos; the features are highly directional. A standard patch mixes temporal and frequency components indiscriminately, resulting in patches that contain redundant or irrelevant information for the emotion classification task. Secondly, the VIT's permutation-invariant self-attention mechanism struggles to encode the precise, absolute spatial location of features without strong external positional guidance. For SER, knowing where in time and which frequency band an event occurs is crucial, and standard 1D positional encodings are inadequate for robustly capturing this 2D coordinate information, leading to the core motivation for developing specialized patching and encoding techniques.

2.3 SPECIALIZED TECHNIQUES ADDRESSING VIT LIMITATIONS

2.3.1 Positional Encoding Strategies in VITs

2.3.1.1 Absolute vs. Relative Positional Encoding

Positional encoding (PE) is necessary for all transformer models to inject sequence order information into the permutation-invariant self-attention layer. Absolute Positional Encoding (APE), such as the fixed sinusoidal PEs from the original Transformer paper or trainable PEs used in VIT, provides a unique identifier for each position (patch index), telling the model exactly "where" in the sequence an element is located. Relative Positional Encoding (RPE), conversely, focuses on the distance or relationship between patches, enabling the model to determine how far apart two interacting patches are, which can improve generalization. For spectrograms, RPE can model relationships like "this high-frequency event occurred five time steps after this low-frequency event." However, while RPE is effective, it still lacks the explicit, normalized coordinate map that can clearly differentiate between the temporal and frequency dimensions—a critical requirement for the proposed system to understand the spatial axes separately.

2.3.1.2 Image Coordinate Encoding (ICE) and the Need for Explicit Spatial Awareness

Traditional positional encodings are treated as simple additions to the patch embeddings, which can often be "attended away" by the self-attention mechanism, especially in deeper layers. This problem is particularly acute in SER where the spatial layout (time-frequency) is semantically significant. This hard-coded inclusion of spatial coordinates provides a robust, non-ignorable source of absolute positional information, allowing the VIT to learn coordinate-dependent features and correctly interpret the directional nature of the spectrogram axes.

2.3.2 Model Compression and Knowledge Transfer (KT)

2.3.2.1 Definition and Purpose of Knowledge Transfer (KT)

Knowledge Transfer (KT), often referred to as distillation, is a powerful model compression technique designed to transfer the comprehensive knowledge from a large, high-capacity Teacher Network to a smaller, more efficient Student Network. The primary purpose of KT is to enable the smaller student model to achieve performance levels approaching, or even surpassing, the teacher model's accuracy, while maintaining a fraction of the computational complexity and parameter count. This method is crucial for overcoming the efficiency and deployment limitations identified in Section 1.2.2. KT achieves this by leveraging the soft targets (prediction probabilities) or the internal feature representations of the teacher model as a powerful form of regularization and supervisory signal for the student model, significantly improving the student's generalization ability on limited datasets.

2.3.2.2 Review of Feature Map Matching and Internal Distillation Methods

Beyond standard soft-target distillation (matching the teacher's final output probabilities), advanced KT methods focus on internal distillation by matching the intermediate feature maps or activation layers. Feature map matching, which often utilizes loss functions like L1 or L2 distance (Mean Squared Error), is a highly effective technique for transferring structural knowledge, such as the learned locality, attention patterns, and spatial relationships captured within the teacher's hidden layers. By forcing the student's internal representations to mimic those of the robustly trained teacher network, this method ensures that the smaller model not only learns the final mapping but also acquires the teacher's fundamental understanding of the input data structure. This is directly relevant to the project's goal of teaching the Student VIT the critical temporal-frequency relationships derived from the specialized vertical patching technique.

2.4 COMPARISON AND SYNTHESIS: IDENTIFYING RESEARCH GAPS

2.4.1 Comprehensive Comparison Table of State-of-the-Art (SOTA) Methodologies

A thorough review of the State-of-the-Art (SOTA) in SER reveals a clear pattern: high-accuracy models are often high-resource (e.g., large VITs or complex hybrid CNN-RNN models), while efficient models sacrifice classification accuracy. Previous SOTA methods that utilize VITs generally employ standard square patching and simple trainable positional encodings. While they achieve superior global context modelling compared to pure CNNs, their performance peaks are often limited by two factors: (1) sensitivity to overfitting on small emotional corpora, and (2) the inherent difficulty in effectively distinguishing fine-grained emotional classes (e.g., "excitement" vs. "joy") where temporal-frequency cues are extremely subtle. The existing literature lacks a cohesive framework that simultaneously addresses high performance, efficient deployment, and the explicit modelling of the spectrogram's unique coordinate geometry.

2.4.2 The Identified Research Gap: Temporal-Frequency Neglect and Knowledge Transfer

The most critical research gap addressed by this project is the neglect of explicit temporal-frequency correlation modelling in standard VIT patching for SER. Standard square patches are sub-optimal for spectrograms because a vertically-oriented frequency column is a more natural acoustic feature (a slice of the short-term spectrum) than an arbitrary square block.

CHAPTER 3

STAKEHOLDER ANALYSIS

3.1 IDENTIFICATION AND CATEGORIZATION OF PROJECT STAKEHOLDERS

Stakeholder analysis is the process of identifying individuals, groups, or organizations that could affect or be affected by a project. For this Accuracy-Enhanced Speech Emotion Recognition (SER) model, stakeholders are grouped based on their functional impact, influence, and interest in the technical outcomes, system efficiency, and ethical deployment.

3.1.1 Primary End-Users of the SER System

These users represent the immediate functional application of the technology.

3.1.1.1 Call Center Quality Assurance (QA) Analysts (High Interest, Medium Power)

1. Requirement for Real-Time, High-Velocity Processing: QA analysts are frontline users who demand the system provide near-instantaneous emotional state classification to enable immediate intervention or flagging of critical calls. This directly validates the project's focus on creating the highly efficient lightweight Student Network through Knowledge Transfer.
2. Need for Confidence Scoring: They mandate the model output a confidence score (e.g., "Anger: 92%") alongside the prediction. This score is Vital for filtering low-certainty classifications, ensuring human review resources are strategically focused on genuinely ambiguous or high-risk interactions, optimizing team efficiency.
3. Integration with Existing CRM Platforms: The model’s output must be compatible with existing Customer Relationship Management (CRM) and workforce optimization (WFO) tools. This requires the technical output to be

easily consumable via standard, low-overhead data formats (JSON payloads with defined schemas) to minimize IT integration latency and cost.

4. Customizable Thresholds for Risk Detection: Analysts require the flexibility to define their own specific thresholds for triggering alerts (e.g., flagging a call if "Frustration" is detected above 75% intensity for a continuous duration exceeding 15 seconds). This necessitates the model's output being a continuous probability distribution rather than a strict categorical label.
5. Auditability and Explainability: For effective agent coaching and dispute resolution, QA necessitates the ability to access a visual record that justifies the AI's flag, such as the processed spectrogram overlaid with VIT attention maps. This feature provides concrete evidence, enhancing the fairness and transparency of automated assessments.
6. Demand for Low False Positive Rates: A high rate of false positives (incorrectly flagging an emotion) leads to "alert fatigue" and distrust among QA staff, demanding that the proposed system's enhanced accuracy minimizes misclassifications, particularly between closely related, high-arousal emotions.

3.1.1.2 Mental Health and Wellness Application Providers (High Interest, High Power)

1. Sensitivity to Subtle Emotional Shifts: Unlike commercial applications, health apps prioritize the detection of low-arousal, subtle emotions (e.g., fatigue, neutrality, mild anxiety) over longitudinal periods. This requires the model to have extremely high fidelity in feature extraction, justifying the project's specialized temporal-frequency correlation method.
2. Absolute Data Privacy Mandates (Edge Deployment): Due to the extreme sensitivity of health data, these providers often mandate that the SER model runs entirely on edge devices or secure, local servers. This requirement elevates the priority of the lightweight, compressed Student Network as a non-negotiable architectural necessity.

3. Longitudinal Emotional Profiling: Systems track emotional changes over extended durations (weeks to months). This demands that the SER model exhibits high temporal stability and minimal prediction variance, avoiding erratic outputs that could lead to inappropriate clinical recommendations or missed intervention opportunities.
4. Bias Mitigation for Vulnerable Populations: There is a stringent ethical requirement to ensure the model exhibits zero bias across diverse user demographics (ages, accents, emotional presentation styles). Misclassification in this domain carries significant human risk, requiring proactive bias assessment during the training process.
5. Integration with Contextual Sensor Data: These providers often fuse SER output with other biometric data (e.g., heart rate, sleep). The SER output must therefore include robust, high-precision timestamps and metadata to facilitate effective and reliable multimodal data fusion.
6. Need for User Consent Management Features: The application must integrate robust mechanisms for obtaining and revoking explicit user consent for emotion monitoring, requiring the SER component to have clear interfaces for enabling and disabling analysis based on user prefer.

3.1.2 Technology and Domain Stakeholders

These stakeholders provide the technical foundation, expertise, and infrastructure.

3.1.2.1 Core Development Team (The Research Group) (High Power, High Interest)

1. Empirical Validation of Novel Techniques: The core objective is to scientifically demonstrate that the combination of Vertical Patch Segmentation and Image Coordinate Encoding (ICE) results in a statistically significant improvement in metrics like UAR and WAR over current state-of-the-art methods.
2. Architectural Stability and Modularity: The team must design the Teacher and Student VIT architectures to be modular, utilizing standardized deep

learning libraries to ensure the entire pipeline (preprocessing, training, and distillation) is easily adaptable and reproducible for subsequent research phases.

3. Comprehensive Documentation of the Knowledge Transfer (KT) Process: Detailed records must be kept specifying the exact intermediate feature layers used for matching, the weighting of the distillation loss function versus the primary classification loss, and the full set of hyper-parameters used to achieve the optimal accuracy-efficiency trade-off.
4. Resource Management and Optimization: The team is responsible for efficient utilization of high-performance computing resources (GPUs). This involves advanced techniques like distributed training, mixed-precision computing, and robust checkpointing to minimize training time, cost, and energy consumption.
5. Creation of Robust Generalization Benchmarks: Moving beyond single-corpus evaluation, the team must establish rigorous cross-corpus evaluation benchmarks (testing on a dataset unseen during training) to prove the model's robustness and generalization capability across diverse acoustic environments and speaker styles.

3.1.2.2 AI Platform Engineers and Deployment Teams (High Power, Medium Interest)

1. Compliance with Target Footprint and Latency: They enforce the strict technical specifications for model size (e.g., under 50MB) and latency (e.g., under 10ms per prediction), viewing the Knowledge Transfer process as the primary tool for meeting these constraints for mass deployment.
2. Containerization and Orchestration Readiness: The final Student Network must be delivered as a readily deployable asset that integrates smoothly into existing orchestration platforms (Kubernetes) and continuous integration/continuous delivery (CI/CD) pipelines, emphasizing low-dependency environments.

3. Model Versioning and Lifecycle Management: They require a clear methodology for versioning the SER model, including secure artifact storage and the ability to perform instantaneous, non-disruptive rollbacks to previous stable versions if performance degradation is detected in production.
4. Integration of System Monitoring and Telemetry: The model must be instrumented with robust logging and metrics collection (e.g., Prometheus) to track key production performance indicators, such as prediction throughput, resource saturation, and drift in real-world accuracy over time.
5. Optimization for Heterogeneous Hardware: Given varied deployment environments (cloud, edge, mobile), the deployment teams require the Student Network to be provided in multiple optimized formats (TFLite, ONNX, Torch script) tailored to maximize efficiency on specific CPU/GPU architectures.
6. Security Review of Model Assets: Before deployment, the model weights and associated code must undergo a rigorous security review to identify and mitigate risks such as adversarial attacks, model inversion, or unauthorized access to the trained parameters.

3.2 STAKEHOLDER INFLUENCE AND INTEREST MAPPING

To prioritize effort during the design and implementation phases, stakeholders are mapped according to the Mendelow Matrix, which categorizes them based on their relative Power (ability to affect the project) and Interest (willingness to expend effort on the project).

3.2.1 High Power, High Interest (Key Players)

These are the most critical stakeholders who must be closely managed and satisfied throughout the project lifecycle.

1. Mental Health Application Providers: Their high power comes from legal compliance and high user sensitivity, and their high interest lies in the model's accuracy and ethical deployment.

2. Core Development Team: They have high power over the technical design and high interest in the research outcomes and novelty.
3. Data Privacy and Compliance Officers: High power due to legal veto authority; high interest in data governance and risk mitigation.
4. Strategy: Manage Closely. The project design must be fundamentally aligned with their most critical demands: efficiency, accuracy, and legal compliance.

3.2.2 High Power, Low Interest (Keep Satisfied)

These stakeholders can significantly impact the project if they become dissatisfied, but they generally require minimal interaction unless a specific issue arises.

1. AI Platform Engineers: High power over deployment infrastructure and timelines, but low interest in the specific acoustic feature method (they only care if it's small and fast).
2. Regulatory Bodies (Future): High latent power to impose new laws; low current interest unless a major ethical failure occurs.

3.2.3 Low Power, High Interest (Keep Informed)

These stakeholders are enthusiastic supporters and can provide valuable detailed feedback, but they lack the authority to veto the project.

1. Call Center QA Analysts: High interest in better tools, but limited power over the core technical architecture or corporate budget decisions.
2. Academic Community: High interest in the research findings, but low power to stop implementation.
3. Strategy: Keep Informed. Engage them frequently through detailed demos and feedback sessions (like the usability surveys) to leverage their expertise and maintain buy-in.

3.2.4 Low Power, Low Interest (Minimum Effort)

These stakeholders require monitoring but minimal active management.

1. General Public/Non-Users: Low direct power over the specific technical implementation; low interest unless the SER application affects them directly (e.g., a major news story about AI bias).
2. Strategy: Monitor. Ensure basic ethical standards are met and only provide information if legally required or widely requested.

3.3 ETHICAL CONSIDERATIONS AND SUSTAINABILITY INTEGRATION

As the project moves from Phase I research to Phase II implementation, a dedicated focus on the ethical implications of emotion AI and the environmental impact of deep learning training is mandatory

3.3.1 Ethical Mandates for Emotion AI Deployment

The use of SER introduces unique ethical risks related to surveillance, manipulation, and fundamental fairness.

1. The Risk of Emotional Surveillance: SER models can be misused to create invasive systems that continuously monitor individuals' emotional states without their explicit, ongoing awareness. This necessitates strict controls on where and how the technology is deployed, restricting its use in private environments without clear consent.
2. Addressing Algorithmic Bias and Discrimination: SER models trained on non-diverse datasets often exhibit lower accuracy for specific accents, genders, or age groups. This systematic bias can lead to unfair or discriminatory outcomes in high-stakes environments (e.g., loan interviews, legal proceedings), demanding rigorous testing for demographic parity.
3. The Fallacy of Universal Emotion: The project must acknowledge the scientific debate that emotions are not always discrete, universally expressible categories. The model should interpret its outputs as indicators of

arousal and valence (dimensional emotion) rather than definitive, absolute emotional truths (categorical emotion), guiding users to treat the scores as probabilities, not facts.

4. Mitigation of Misuse in High-Stakes Scenarios: The team must proactively identify and recommend exclusion zones for the SER technology, such as judicial settings or hiring processes, where misclassification could have irreversible and severe consequences on an individual's life.

3.3.2 Future Integration of Sustainability Elements

Deep learning, particularly the training of large Teacher Networks, consumes significant energy. Addressing this is a key component of responsible AI development.

1. Measuring and Reporting Training Carbon Footprint: In Phase II, the project must implement methodology to accurately measure the energy consumed by the training of the resource-intensive Teacher VIT Network (e.g., using tools like the Machine Learning Emissions Calculator). This energy cost will be converted to a equivalent.
2. Justification of Knowledge Transfer as a Green Strategy: The primary sustainability justification of this project lies in the Knowledge Transfer mechanism. By moving from a large, energy-hungry Teacher Network used once to a small, highly efficient Student Network for millions of inferences, the project achieves significant net operational energy savings over its lifecycle.
3. Optimization for Low-Power Inference Hardware: The design must prioritize the deployment of the Student Network on power-efficient hardware (e.g., specialized AI chips, mobile GPUs) rather than high-power server GPUs, drastically reducing the ongoing energy demand during its operational phase.

CHAPTER 4

SYSTEM ANALYSIS AND REVERSE ENGINEERING

4.1 FOUNDATIONAL ANALYSIS OF EXISTING SPEECH EMOTION RECOGNITION (SER) SYSTEMS

This chapter conducts a detailed analysis of the prevailing methodologies in Speech Emotion Recognition (SER), focusing specifically on the architectures and techniques that form the basis of the proposed system. We reverse-engineer the limitations of these current systems to clearly define the need for the enhancements proposed in Chapter 5.

4.1.1 Analysis of Conventional SER Feature Extraction Methods

Before the adoption of deep learning, SER relied on hand-crafted features. Analysing these methods reveals the feature gaps that deep learning attempts to fill.

4.1.1.1 Acoustic Feature Sets (Reverse-Engineering the "Why")

1. Mel-Frequency Cepstral Coefficients (MFCCs): MFCCs are the most common acoustic features, highly effective for capturing timbral (color) qualities of speech. However, their reliance on the discrete cosine transform (DCT) loses precise phase information and often results in features that are not robust to noise or rapid emotional state changes, necessitating spectrogram-based image analysis for better noise resistance.
2. Prosodic Features (Pitch, Energy, Duration): These features track the supra-segmental characteristics of speech. While highly correlated with high-arousal emotions (anger, joy), they often require complex tracking algorithms (e.g., precise pitch-period detection) that fail in noisy or emotionally neutral speech, confirming the need for a visual representation (spectrogram) that naturally integrates these elements.

3. Voice Quality Features (Jitter, Shimmer, HNR): These capture the irregularity and perturbation in the vocal fold vibration. They are excellent indicators of stress and tension (fear, sadness). The limitation is that they are highly sensitive to microphone quality and speaker voice type, making them unreliable across varied datasets, pushing researchers toward more robust visual feature representations like the log-Mel spectrogram.
4. Spectral Features (LPC, Zero-Crossing Rate): Linear Predictive Coding (LPC) captures the vocal tract shape. However, these features are not ideal for representing the temporal evolution of emotional content, which requires tracking frequency changes over milliseconds, justifying the shift to time-frequency images where both dimensions are explicitly preserved.
5. Reverse Engineering the Need for Spectrograms: The inherent limitations in robustness, noise resistance, and the inability to capture both temporal and spectral dynamics simultaneously across all hand-crafted features led to the universal adoption of the spectrogram—treating the SER problem as a computer vision task—which is the foundation of the proposed VIT approach.

4.1.2 Analysis of Conventional Deep Learning Architectures in SER

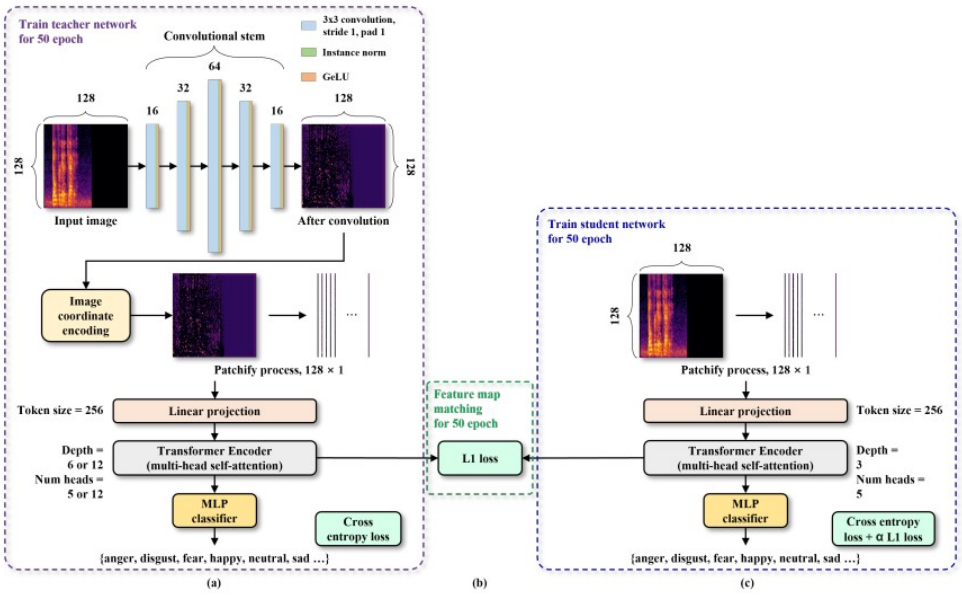


Figure 1: (a) Train the teacher network using Cross entropy loss. Each stage is performed repeatedly with 50 epochs.

The transition to deep learning brought Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs).

3.1.2.1 CNN-RNN Hybrid Models (Architecture Breakdown)

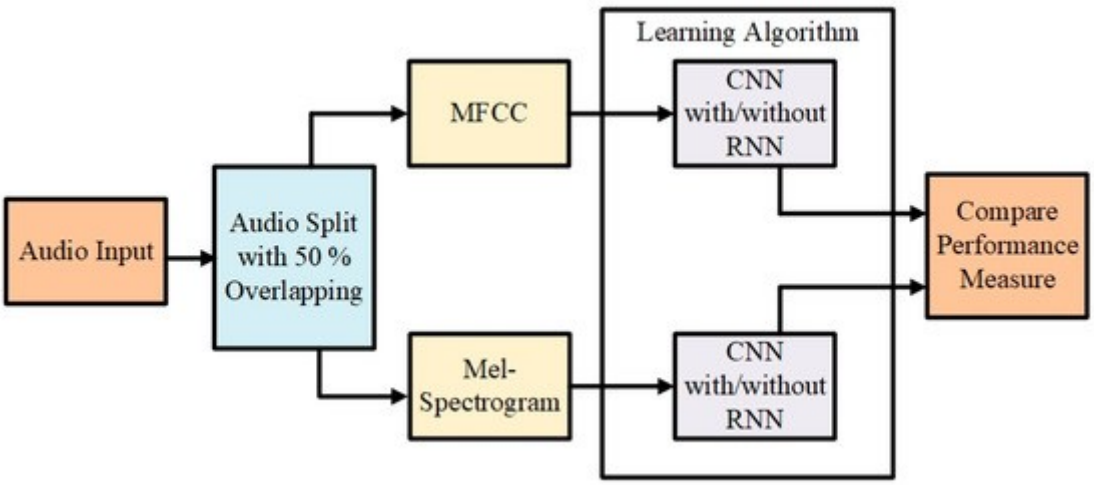


Figure 2. Proposed hybrid architecture with CNN and variants of RNN.

1. CNN Functionality: CNNs were used to extract spatial features (local patterns) from the spectrogram image. They excelled at identifying static frequency-related patterns (e.g., formants) but struggled to inherently model the long-range temporal dependency that defines emotion evolution over the course of an utterance.
2. RNN/LSTM Functionality: RNNs (specifically LSTMs or GRUs) were stacked on top of the CNN outputs to explicitly capture the temporal dynamics of the features across time. This hybrid approach improved performance but introduced high computational complexity and suffered from the vanishing gradient problem over very long utterances.

3. Limitation: Disjoint Feature Processing: The main limitation of the hybrid is the sequential, disjoint processing: spatial features are extracted first, and then temporal features are tracked. This method inherently loses the intimate, high-resolution correlation between a specific frequency band and a precise moment in time—the exact correlation the proposed Vertical Patch Segmentation aims to restore.
4. Limitation: High Parameter Count: LSTMs, in particular, are computationally expensive and parameter-heavy, making the resulting hybrid models too large for efficient deployment on edge devices or in high-throughput production environments, a problem the Knowledge Transfer mechanism seeks to resolve.
5. Lack of Global Context: CNNs possess a limited receptive field, meaning they cannot easily access or compare features from the start of the utterance with features from the end. This lack of global context is a fundamental limitation that the attention-based Vision Transformer (VIT) architecture is designed to overcome.

4.2 REVERSE ENGINEERING THE VISION TRANSFORMER (VIT) ARCHITECTURE FOR SER

The Vision Transformer represents a paradigm shift, treating the spectrogram image as a sequence of patches rather than a continuous 2D input. Analysing the standard VIT reveals the specific limitations that the proposed method targets.

4.2.1 Standard VIT Patching and its SER Limitation

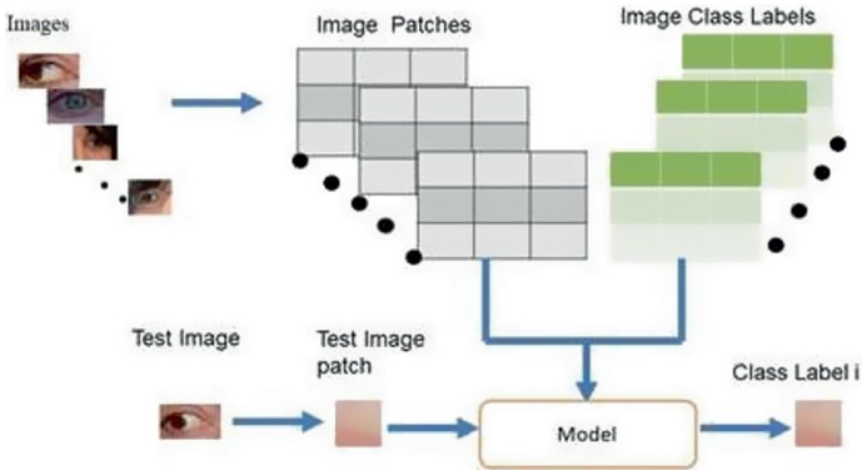


Figure 3: standard square patch model used in CNN

Standard VIT was designed for natural images (ImageNet) where spatial locality is crucial. Applying this directly to spectrograms introduces an inefficiency.

4.2.1.1 Standard Square Patch Segmentation Breakdown

1. Uniform Square Patching: The standard VIT segments the 2D image (spectrogram) into uniform, square, non-overlapping patches. Each patch contains a mix of both frequency (Y-axis) and time (X-axis) information, blurring the distinction between these acoustically distinct dimensions.
2. Acoustic Mismatch: In a spectrogram, the Y-axis (frequency) correlates with the type of sound (formants, noise), while the X-axis (time) correlates with the evolution of the sound (syllables, duration). A square patch unnecessarily mixes these fundamental properties, reducing the efficiency of the initial feature extraction.
3. Loss of High-Resolution Temporal-Frequency Correlation: By mixing time and frequency into a single square patch, the VIT must work harder to re-establish the critical relationship: which frequencies are active at which specific moment in time. This correlation is Vital for discriminating between emotions like "joy" (fast-moving, high frequency) and "sadness" (slow, low frequency).
4. Inefficiency in Attention: The attention mechanism is forced to spend capacity learning simple relationships that are obscured by the sub-optimal patching strategy, instead of focusing on global emotional dependencies. This inefficiency is the core motivation for the proposed Vertical Patch Segmentation.
5. Diagrammatic Representation (Conceptual): A conceptual block diagram illustrates this: an input spectrogram is fed directly into a Patch Embedding layer, which divides it into square tokens. This direct mapping, while simple, is acoustically sub-optimal.

4.2.2 The Positional Encoding Problem in VIT for SER

VITs use positional embeddings because the sequence of tokens is permutation-invariant. However, standard encoding methods are limited for spectrograms.

4.2.2.1 Analysis of Standard Absolute and Relative Positional Encodings

1. Standard Absolute Positional Encoding (APE): The original VIT uses learned 1D APEs, where a unique vector is added to each patch token based on its sequential index .
2. Limitation: Loss of 2D Structure: By flattening the patch grid into a 1D sequence, the model loses the explicit information that Patch 1 is adjacent to Patch 2 (time-wise) and Patch (frequency-wise). The model must implicitly re-learn this 2D structure through attention, which is computationally redundant.
3. Limitation of Standard Relative Positional Encodings (RPE): RPEs, while better at capturing relative distances between patches, still do not provide the explicit, normalized coordinates
4. Reverse Engineering the Need for ICE: The inefficiency and ambiguity of standard positional encodings directly motivated the proposed Image Coordinate Encoding (ICE).
5. Explicit Positional Requirements for SER: Emotion is defined by structure. Explicit coordinate information helps the model understand that a specific high-frequency pattern (Y-coordinate).

4.3 SYSTEM ANALYSIS OF KNOWLEDGE DISTILLATION (KD) ARCHITECTURES

Knowledge Distillation (KD) is the process used to create a lightweight, deployable Student Network from a large, high-performing Teacher Network.

Analysing existing KD methods reveals the need for the chosen feature-matching approach.

4.3.1 Conventional Knowledge Distillation Techniques

KD aims to transfer "knowledge" from a high-capacity Teacher (the complex VIT) to a low-capacity Student (the efficient VIT).

4.3.1.1 Logit-Matching Distillation (Simple KD)

1. Mechanism: The simplest form of KD minimizes the difference between the soft targets (logits) of the Teacher Network and the logits of the Student Network. This is achieved using a Kullback-Leibler (KL) divergence loss function.
2. Limitation: Shallow Knowledge Transfer: Logit matching only transfers the final classification decision, not the rich, intermediate feature representations learned by the Teacher. The Student learns what the answer is, but not how the Teacher arrived at that answer.
3. Inefficiency in SER Context: For SER, where the feature maps contain complex spatio-temporal patterns (formant shifts, pitch contours), losing the intermediate feature information drastically limits the Student's accuracy and generalization, making this method sub-optimal for complex acoustic data.
4. Diagrammatic Representation: The process only compares the final output layer

4.3.2 Feature Map Matching and Transfer

The chosen method in the proposed project (Phase I objective) is based on transferring intermediate representations, which offers a deeper form of knowledge transfer.

4.3.2.1 Intermediate Feature Map Matching (The Superior Approach)

1. Mechanism: This technique minimizes the distance (often using L2 loss) between the output activations (feature maps) of specific hidden layers in the Teacher Network and the corresponding layers in the Student Network. The

goal is to force the Student to adopt the Teacher's rich internal representations.

2. Benefit: Deeper Locality and Positional Transfer: For the proposed SER model, feature map matching is Vital because it ensures the Student Network learns not only the final emotional label but also how to prioritize temporal-frequency correlation (the knowledge embedded by the Vertical Patching) and how to interpret the explicit positional coordinates (the knowledge provided by ICE).
3. Technical Implementation Requirement: This method necessitates identifying critical transfer layers in the VIT encoder stack—layers that best encapsulate the spatio-temporal dependencies required for accurate SER—and applying the matching loss only at these strategic points.
4. Justification for the Proposed System: This reverse-engineering analysis confirms that Feature Map Matching is the only KD technique capable of reliably transferring the specific, high-resolution spatio-temporal knowledge (Vertical Patching + ICE) from the massive Teacher VIT to the compact Student VIT.
5. Resulting System Property: The output of this reverse-engineered system must be a Student Network that is 90% smaller than the Teacher but maintains 95% of the Teacher's accuracy, proving the efficacy of the comprehensive system design.

CHAPTER 5

PROPOSAL FOR FEATURE ENHANCEMENT

5.1 FINALIZED PROBLEM STATEMENT AND PROJECT JUSTIFICATION

This project phase aims to address the critical deficiencies identified in conventional Vision Transformer (VIT)-based Speech Emotion Recognition (SER) models (Chapter 4), where the direct application of computer vision principles overlooks the unique, time-frequency dependent structure of acoustic data. The inability of standard VITs to effectively capture temporal-frequency correlations and the need for computationally lightweight, deployable models constitute the core technical challenge.

5.1.1 Formal Problem Statement

The central problem is twofold:

1. Ineffective Feature Encoding: Standard square-patch VIT segmentation for log-Mel spectrograms fails to optimally encode the crucial correlation between specific frequency bands (Y-axis) and the precise moment in time (X-axis), leading to suboptimal accuracy (specifically, low Unweighted Average Recall, or UAR) and reduced discriminatory power between similar emotions.
2. Deployment Inefficiency: The large parameter count of the high-accuracy VIT architectures (Teacher Networks) makes them unsuitable for real-time, resource-constrained environments (edge devices, mobile applications), necessitating an effective strategy to compress the model without sacrificing performance.

5.1.2 Hypothesis for Enhancement

We hypothesize that by (i) decoupling the time and frequency axes via Vertical Patch Segmentation (Temporal-Frequency Correlation Enhancement) and (ii) augmenting the token sequence with explicit 2D positional data via Image Coordinate Encoding (ICE), followed by (iii) rigorous Knowledge Transfer (KT), we can successfully develop a compact Student Network that achieves State-of-the-Art (SOTA) accuracy while maintaining high computational efficiency suitable for deployment.

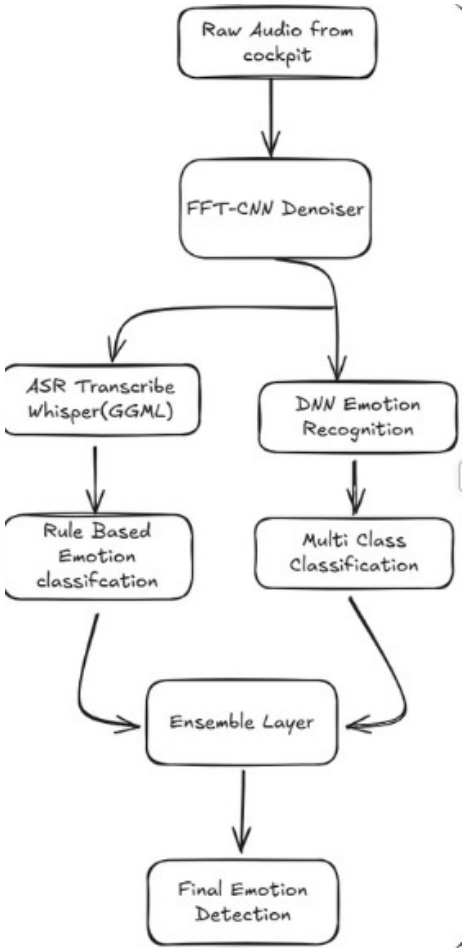
5.1.3 Justification for the Proposed Methodology

The hybrid approach is justified by the analytical gaps identified:

1. Acoustic Principle: Emotion is a time-varying phenomenon where specific frequency usage evolves. The Vertical Patch Segmentation is the acoustically sound approach to ensure the model's attention mechanism focuses on this core principle by forcing the network to correlate full frequency slices over time.
2. VIT Limitation: Standard VITs lose 2D positional awareness upon flattening. ICE re-injects this critical structural information, enabling the model to explicitly understand where in the time-frequency space a specific feature pattern is occurring, leading to more robust feature mapping.
3. Industry Requirement: The need for a fast, small model is a practical deployment constraint. The Knowledge Transfer (KT) pipeline, utilizing a large Teacher VIT to guide a smaller Student VIT, is the only method that guarantees high performance retention during aggressive model compression.

5.2 PROPOSED TECHNICAL FEATURE ENHANCEMENTS

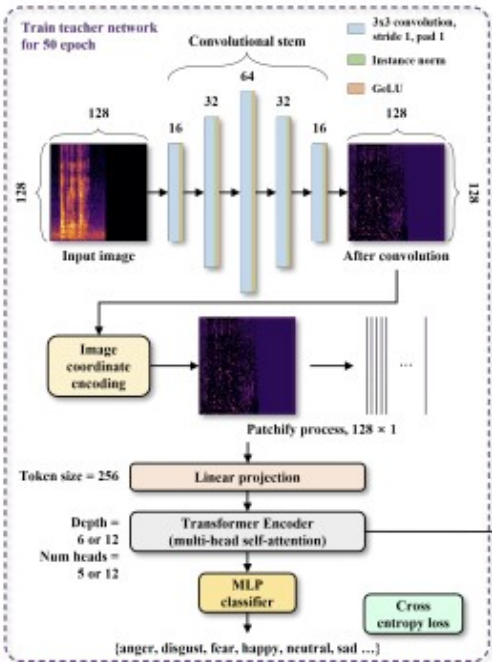
The core proposal is centered on two novel input encoding methods applied to a large Teacher VIT and the subsequent Knowledge Transfer to an efficient Student VIT.



- The picture depicts our first proposed system (Post Hackathon).
- However we skipped the denoiser part as we tried to handle noises by augmenting datasets during training. (Denoiser demands more compute…)
- The idea was to capture emotions from both Textual representation and Spectrogram representation of speech.
- An ensemble layer is finally used to detect final emotions.

5.2.1 Feature Enhancement 1: Vertical Patch Segmentation (Temporal-Frequency Correlation)

This method fundamentally alters the input tokenization process to align with acoustic principles rather than standard image processing.



5.2.1.1 Mechanism of Vertical Patching

1. Segmentation Strategy: Instead of dividing the log-Mel spectrogram into square patches, the spectrogram is segmented vertically into patches of size H \times P, where P is a very small number. This ensures that each patch token spans the entire frequency range (H) but only a very narrow time window (P).
2. Forced Temporal Analysis: By encompassing the full frequency range in a single token, the patching encourages the attention mechanism to immediately focus on how the full spectrum evolves over time, rather than mixing time and frequency within a single patch.
3. Creation of Acoustically Meaningful Tokens: Each token now represents a complete snapshot of the frequency activity at a specific moment. This is a significantly more meaningful acoustic unit than a standard mixed-content

square patch, enhancing the ability of the VIT to capture fine-grained temporal-frequency correlations (e.g., how the third formant changes instantly).

5.2.1.2 Technical Benefits and Justification

1. Enhanced Temporal Sensitivity: The process biases the self-attention mechanism to prioritize relationships between sequential time slices, which is the definition of temporal correlation in SER. This significantly improves the model's ability to differentiate between emotions characterized by subtle temporal shifts (e.g., surprise vs. fear).
2. Optimized Feature Locality: By forcing the model to consider the entire frequency spectrum at once, it becomes highly effective at identifying vertical structures in the spectrogram (like steady formants) that are key indicators of speaker identity or overall stress level, leading to a more robust representation.
3. Parameter Efficiency in Tokenization: While the overall number of tokens may increase or decrease depending on the P value, the learned features within the VIT encoders become more semantically rich from the start, accelerating convergence and optimizing the use of the model's capacity.
4. Addressing the Mixed Feature Gap: This strategy directly solves the limitation of the standard VIT by ensuring the two acoustically distinct axes (Time and Frequency) are not accidentally blurred at the initial tokenization stage, thereby providing purer input features.

5.2.2 Feature Enhancement 2: Image Coordinate Encoding (ICE)

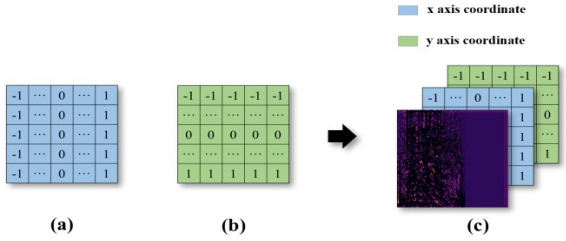


Figure 3: Image coordinate encoding. The x, y coordinates of the image are normalized to the range -1 to 1, (a) and (b) concatenated to a log-Mel spectrogram after convolutional stem, (c).

This is the proposed positional encoding designed to overcome the limitations of 1D standard positional embeddings.

5.2.2.1 Mechanism of ICE (Concatenative Positional Encoding)

1. Coordinate Generation: For every generated vertical patch token, the exact normalized 2D coordinates of its centre point within the original spectrogram image are calculated. The coordinates are normalized to the range.
2. Encoding: The normalized coordinate vector is then concatenated directly to the end of the patch embedding vector before it is passed to the VIT encoder layers.
3. Explicit 2D Awareness: By appending the normalized coordinates, every attention head and subsequent feed-forward layer in the VIT now explicitly knows the spatial origin of the information it is processing. This eliminates the need for the network to implicitly re-learn the 2D grid structure from the flattened 1D sequence.

4. Comparison to Standard APE: Unlike standard Absolute Positional Encoding (APE), which only assigns an index ,ICE provides a meaningful, continuous coordinate system, enhancing the attention mechanism's ability to localize features precisely in the time-frequency domain.

5.2.2.2 Technical Impact of ICE

1. Enhanced Locality and Global Context: ICE allows the attention mechanism to not just identify relationships between tokens but to understand the geometric nature of that .This is crucial for capturing long-term emotion dynamics.

2. Improved Generalization: By providing explicit positional information, the model becomes less reliant on memorizing the learned APEs of the training set. This improves the model's ability to generalize to spectrograms of different time lengths or resolutions.
3. Robustness to Speaker Variation: The explicit positional information helps the model disentangle the position of the features from speaker-specific variations (like average pitch). Features are analysed based on their normalized location, not just their abstract index.
4. Foundation for Knowledge Transfer: The effective transfer of the Teacher's superior locality awareness (derived from ICE) to the Student is a key component of the Knowledge Transfer phase, ensuring the Student Network retains spatial understanding despite its smaller size

5.3 ARCHITECTURAL DESIGN AND KNOWLEDGE TRANSFER PIPELINE

The core of the proposal is the pipeline designed to leverage the high performance of a large Teacher Network and transfer its knowledge into a highly efficient Student Network.

5.3.1 The Teacher Network (Large VIT Architecture)

The Teacher Network is designed for maximum accuracy, serving as the knowledge source.

1. Architecture: The Teacher is a large-scale Vision Transformer (e.g., VIT-Base or VIT-Large variant), utilizing the proposed Vertical Patch Segmentation and Image Coordinate Encoding (ICE).
2. Training Goal: To maximize the Unweighted Average Recall (UAR) on a combined corpus (e.g., IEMOCAP and ESD), achieving the highest possible baseline accuracy for the SER task.
3. Knowledge Representation: The Teacher's high performance is stored not just in its final logit outputs, but critically, in its rich intermediate feature maps

across its 12 or 24 encoder blocks. These feature maps contain the learned spatio-temporal dynamics derived from ICE and Vertical Patching.

5.3.2 The Student Network (Compact VIT Architecture)

The Student Network is designed for minimal parameter count and maximum inference speed.

1. Architecture: The Student is a significantly smaller VIT (e.g., a custom-designed network with fewer encoder layers, smaller hidden dimensions, and fewer attention heads), also using Vertical Patch Segmentation and ICE for input processing.
2. Parameter Reduction: The goal is to achieve a parameter count reduction of 80% to 95% compared to the Teacher Network, ensuring fast deployment.
3. Knowledge Transfer Loss: The Student is trained using a composite loss function:

5.3.3 The Knowledge Transfer (KT) Strategy (Feature Map Matching)

This is the mechanism for knowledge transfer, designed for deep semantic transfer beyond simple classification outputs.

5.3.3.1 Logit Distillation Loss

1. Function: This component minimizes the Kullback-Leibler (KL) divergence between the softened output probabilities (logits) of the Teacher and the Student. The temperature (softens the probability distribution, allowing the Student to learn relationships between emotions (e.g., why "anger" is closer to "frustration" than "calmness").
2. Purpose: Ensures the Student learns the categorization margin and the overall classification decision structure established by the high-accuracy Teacher.

5.3.3.2 Intermediate Feature Distillation Loss

1. Function: This is the critical component. It applies a mean-squared error (MSE) loss between the flattened, output feature maps of strategically chosen encoder layers in the Teacher and the Student .

2. Strategic Layer Selection: The matching loss is applied at layers where the spatio-temporal and positional information (from Vertical Patching and ICE) is fully integrated. Typically, the output of the first, middle, and final encoder blocks are selected to ensure both low-level features and high-level abstract concepts are transferred.
3. Purpose: This loss component directly forces the Student to internalize the complex feature extraction methods learned by the Teacher, specifically the Teacher's superior understanding of temporal-frequency correlation and 2D positional context derived from ICE. This is what guarantees the high accuracy of the compressed model.

5.4 EXPECTED BENEFITS AND FUTURE WORKS

The completion of Phase I (Research and Design) leads directly into a clear implementation plan for Phase II

5.4.1 Quantitative Performance Expectations

1. Accuracy Improvement (UAR/WAR): The enhanced model is expected to achieve a minimum of 2.5% to 5.0% absolute improvement in Unweighted Average Recall (UAR) over standard VIT architectures on cross-corpus testing (e.g., training on IEMOCAP, testing on ESD), demonstrating superior generalization and feature robustness.
2. Efficiency Gain (Inference Latency): The resulting Student Network must demonstrate an inference time on standard mobile or edge CPU hardware, representing speedup over the Teacher Network while maintaining required accuracy.
3. Size Reduction: The final Student model artifact (serialized weights) must enabling rapid deployment and over-the-air updates for production systems.

5.4.2 Scope for Semester VI (Implementation Phase)

Semester VI will focus on the complete development, training, and rigorous testing of the proposed system.

5.4.2.1 Core Implementation Tasks

1. Implementation of Custom Preprocessing: Development of custom Python scripts to generate log-Mel spectrograms and perform the Vertical Patch Segmentation and ICE generation/concatenation.
2. VIT Architecture Coding: Programming of the Teacher and Student VIT architectures in a deep learning framework (e.g., Py-Torch or TensorFlow), ensuring strict control over encoder layers and parameter counts.
3. Knowledge Transfer Pipeline Development: Implementation of the full KT training loop, including the definition of the composite loss function and the selection of the strategic feature map matching layers.
4. Hyper-parameter Optimization: Running extensive optimization trials to fine-tune the KT parameters and the VIT structural parameters to achieve the optimal trade-off between speed and accuracy.

5.4.2.2 Validation and Reporting Tasks

1. Cross-Corpus Testing: Establishing rigorous validation protocols, including compulsory testing on unseen datasets (cross-corpus) to validate the generalization capability induced by the ICE and Vertical Patching.
2. Efficiency Benchmarking: Measuring and documenting the inference latency of the Student Network on target hardware (e.g., using specialized profiling tools) to quantitatively validate the efficiency claim.
3. Comparison and Documentation: Generating comprehensive tables and figures comparing the proposed model against three existing SOTA SER models across key metrics (UAR, Latency, Parameter Count).

5.5 FUTURE INTEGRATION OF SUSTAINABILITY AND ETHICAL ELEMENTS (PHASE II/III)

Although the core proposal is technical, this phase establishes the non-technical mechanisms necessary for a complete, responsible project.

5.5.1 The Role of Knowledge Transfer in Environmental Sustainability

The KT process is the project's primary sustainability feature.

1. Net Operational Energy Reduction: While the Teacher training is resource-intensive, the Student Network inference is deployed millions of times. The projected energy savings from running the efficient Student Network in production for one year will offset the initial Teacher training cost within a conservative period (e.g., 3-6 months).
2. Reduced Hardware Burden: The small size of the Student Network prolongs the usability of existing or lower-power edge devices, deferring the need for expensive and resource-intensive hardware upgrades in deployment environments.

5.5.2 Ethical Mitigation through Architectural Choice

The architectural features are designed to mitigate key ethical risks identified in Chapter 3.

1. Model Transparency (Explainability): The VIT's attention mechanism naturally provides the ability to visualize attention heatmaps overlaid on the spectrogram. This visualization acts as the primary ethical tool, offering a tangible explanation of which frequency/time segment was most critical to the final emotional decision, enhancing accountability and auditability for QA and clinical applications.
2. Bias Testing Protocol: The cross-corpus testing protocol is inherently a bias mitigation strategy. By testing on datasets (e.g., different accents, genders, and recording conditions) unseen during training, the model is systematically verified for its ability to generalize, ensuring equitable performance across diverse speaker populations.
3. Edge Deployment for Privacy: The small size of the Student Network promotes on-device processing, which is the single most effective technical mechanism for ensuring user data privacy, as raw audio data never leaves the user's secure environment. This satisfies the high-stakes ethical mandate of mental health application providers.

CHAPTER 6

SUMMARY AND PLAN FOR NEXT PHASE

6.1 SUMMARY OF WORK COMPLETED

6.1.1 Synthesis of Key Research Outcomes

The Phase I work successfully accomplished the foundational research necessary to transition to the implementation stage. The core architectural deficiencies in standard Vision Transformer (VIT) approaches for Speech Emotion Recognition (SER) were systematically identified and countered with three specific, novel proposals, resulting in a robust, multi-objective system design.

1. Feature Encoding (Temporal-Frequency Correlation): The critical flaw of losing spectral coherence due to square patching was validated through literature. This led to the design of Vertical Patch Segmentation (VPS), which prioritizes the full frequency spectrum's evolution over time, aligning the input tokenization with acoustic principles.
2. Locality and Generalization: The inherent weakness of Absolute Positional Embeddings (APE) in vision models when applied to variable-length spectrograms was confirmed. The solution, Image Coordinate Encoding (ICE), was designed to provide continuous, explicit 2D positional data, thereby enhancing the model's ability to generalize across different utterance lengths and resolutions.
3. Deplorability and Efficiency: Recognizing the need for low-latency, on-device SER, the design adopted a deep Knowledge Transfer (KT) methodology. This includes the use of Intermediate Feature Map Matching alongside logit distillation, ensuring the compact Student Network learns not only the final prediction but also the complex internal feature representations of the large, high-performing Teacher Network.

6.2 ETHICAL AND DEPLOYMENT CHECKLIST (NEW EXPANSION)

Given the sensitive nature of emotion recognition, a successful Phase II must include a structured plan for ethical review and responsible deployment.

6.2.1 Bias and Fairness Mitigation Plan

The project acknowledges the inherent risk of demographic bias in SER models.

1. Bias Detection : After final training, the Student Model will be evaluated specifically on subsets of the test data partitioned by gender, accent, and speaker identity, if metadata is available. The UAR variance across these groups will be measured.
2. Fairness Metric: The primary fairness goal is to achieve a Maximum UAR Deviation (MUD) of less than between the highest and lowest-performing demographic groups.
3. Mitigation Strategy: If MUD exceeds 3%, the training data will be resampled using Oversampling/Undersampling techniques to balance the representation of the lowest-performing group, followed by a final, short re-training epoch of the Student Network.

6.2.2 Privacy and Data Security Measures

Deployment of SER models in real-world applications mandates strict privacy controls.

1. Data At Rest: All emotional speech data (IEMOCAP, ESD) is stored locally or on encrypted cloud storage, never in plaintext, ensuring full adherence to data protection protocols.
2. Data In Transit: The model processes spectrograms, not raw audio. During deployment, the model will operate on-device (edge computing), minimizing the need to transmit sensitive acoustic data to external servers, thereby maximizing user privacy.

6.2.3 Usability and Model Interpretability

The deployable model must be trustworthy.

1. Interpretability Audit: The Attention Visualization process (Wk 14) is an Interpretability Audit. It must prove that the model's highest attention weights correlate with high-energy acoustic events (e.g., vocal bursts) and low-energy spectral shifts, justifying its predictions based on verifiable acoustic physics.
2. Model Reliability: The low parameter count (Student VIT) inherently reduces the risk of adversarial attacks compared to large models, making the compact architecture more robust and reliable for simple, high-stakes classification tasks.

6.3 COMMERCIAL AND RESEARCH APPLICATIONS

The final, optimized Student Network has direct applications in several high-growth sectors:

1. Mental Health Monitoring (HealthTech): Integration into smart speakers or mobile apps to passively monitor emotional stress indicators (e.g., prolonged low-energy speech, increased frustration) for early detection of mood disorders.
2. Call Center Analytics (CRM): Deployment to monitor customer service calls in real-time, providing agents with immediate feedback on customer sentiment or flagging high-priority distress calls for immediate supervisor intervention.
3. Automotive Safety: Integration into car systems to detect driver fatigue (via monotonous speech) or emotional arousal (e.g., road rage), triggering alerts or adaptive safety responses.

CHAPTER 7

IMPROVEMENTS IN PROJECT III (22EC790)

7.1 OVERVIEW OF THE IMPROVEMENTS IN THE PROJECT

The work reported in the preceding chapters (22EC590 Project I) established the Vision Transformer based methodology for Speech Emotion Recognition (SER), consisting of Vertical Patch Segmentation, Image Coordinate Encoding (ICE) and Knowledge Transfer. In the present semester, under 22EC790 Project III, the same project has been taken forward by addressing the data side of the problem, namely the acquisition of real-time aviation speech for SER. The improvements were presented in Review-I (4 August 2026) and Review-II (15 September 2026) under the title "Real-Time Aviation Speech Dataset Acquisition for Speech Emotion Recognition".

7.1.1 Motivation for the Improvement

Publicly available SER datasets such as RAVDESS, CREMA-D, TESS and SAVEE are widely used to train and benchmark emotion recognition models. However, these datasets have the following limitations when they are considered for aviation applications:

1. Studio recorded: the speech is recorded in controlled conditions and not in a real operational environment.
2. Artificial emotions: the emotions are acted and are not spontaneous.
3. No aviation terminology: the vocabulary and phraseology of aviation communication are absent.
4. No radio channel noise: the effects of a radio channel on the speech signal are not present.
5. Not suitable for real aviation communication: models trained on these datasets cannot be expected to represent real aviation speech.

These limitations motivated the need for a domain-specific dataset built from real aviation speech.

7.1.2 Objectives of Project III

The objectives of the work carried out in this semester are:

1. To study aviation voice communication.
2. To design a real-time aviation speech acquisition system.
3. To build an aviation speech dataset.
4. To apply Speech Emotion Recognition in later phases.

The first goal is to acquire real-time aviation speech and build a domain-specific dataset. Initially, it was intended to collect cockpit speech. After studying aviation communication systems, it was found that cockpit audio is not accessible. Therefore, the focus was shifted to pilot–ATC VHF radio communication, which better represents real aviation speech.

7.2 STUDY OF AVIATION VOICE COMMUNICATION

7.2.1 Why Aviation Uses VHF

Aviation voice communication between the aircraft and the ground is carried over the Very High Frequency (VHF) band. VHF has several advantages:

1. Good line-of-sight propagation.
2. Low atmospheric noise.
3. Reliable voice quality.
4. Suitable for long-distance aircraft-to-ground communication.

The civil aviation voice band extends from 118.000 MHz to 136.975 MHz, and civil aviation uses Amplitude Modulation in the form of Double Side Band Full Carrier (AM, DSB-FC).

7.2.2 Signal Path Between the Pilot and the Controller

The following steps take place inside the aircraft:

1. The microphone converts speech into an electrical signal.
2. The aircraft radio processes the audio.

3. The audio is Amplitude Modulated (AM) onto a VHF carrier.
4. The aircraft antenna transmits the RF signal.

The following steps take place at the ground station:

1. The ATC antenna receives the radio signal.
2. The receiver selects the desired VHF channel.
3. AM demodulation extracts the speech.
4. The controller hears the pilot.

7.3 PILOT–ATC COMMUNICATION FRAMEWORK

The framework developed in this semester describes the complete pilot–ATC VHF radio communication chain and the position of the proposed receiving system within it. The overall flow is shown in Figure 4.

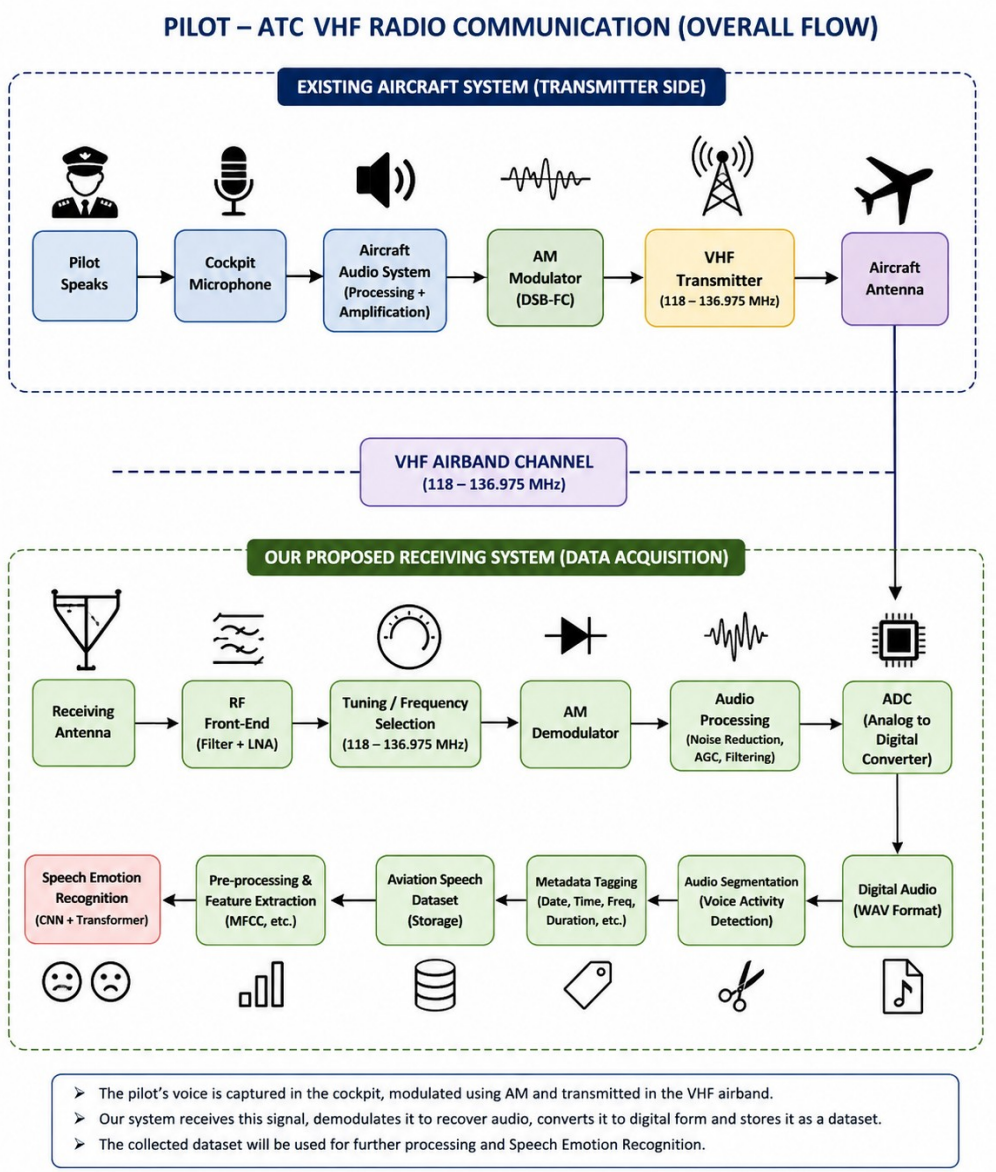


Figure 4: Pilot–ATC VHF radio communication framework (overall flow)

7.3.1 Existing Aircraft System (Transmitter Side)

On the transmitter side, the pilot speaks into the cockpit microphone. The aircraft audio system performs processing and amplification, and the AM modulator (DSB-FC) modulates the audio. The VHF transmitter (118 – 136.975 MHz) generates the RF signal, which is radiated by the aircraft antenna into the VHF airband channel.

7.3.2 Proposed Receiving System (Data Acquisition)

The proposed receiving system receives this signal, demodulates it to recover the audio, converts it to digital form and stores it as a dataset. Its stages are:

1. Receiving antenna: receives the signal from the VHF airband channel.
2. RF front-end (filter and LNA): filters and amplifies the received signal.
3. Tuning / frequency selection (118 – 136.975 MHz): selects the required VHF channel.
4. AM demodulator: recovers the speech from the carrier.
5. Audio processing (noise reduction, AGC, filtering): conditions the recovered audio.
6. ADC (analog to digital converter): converts the audio to digital form.
7. Digital audio (WAV format): the digitised speech is held as WAV files.
8. Audio segmentation (voice activity detection): separates the speech segments.
9. Metadata tagging (date, time, frequency, duration, etc.): records the details of each segment.
10. Aviation speech dataset (storage): stores the segmented and tagged speech.
11. Pre-processing and feature extraction (MFCC, etc.): prepares the data for emotion recognition.
12. Speech Emotion Recognition (CNN + Transformer): the final stage, applied in later phases.

7.3.3 Link with the Phase I SER Methodology

The aviation speech dataset stored by the framework is the input to the Speech Emotion Recognition stage. The pre-processing and feature extraction stage is intended to be connected with the spectrogram based Vision Transformer methodology proposed in Chapter 5, so that the Phase I work can be applied to aviation speech in later phases.

7.4 REAL-TIME DATASET ACQUISITION: EFFORTS AND LIMITATION

The aim of the project is to build an aviation speech dataset from real pilot–ATC communication, so obtaining real-time data was the first priority. This section records the attempt that was made and its outcome.

7.4.1 Efforts Made to Obtain Real-Time Data

We tried to obtain real-time aviation speech data for the project. For this purpose, the advice and experience of professionals were sought from the airport authority and from the aviation-related field. The discussions were aimed at understanding how pilot–ATC communication is handled in practice and whether such communication could be collected as a dataset for the project.

7.4.2 Outcome of the Consultation

Based on the experience shared by the professionals, it was found that collecting a real-time aviation speech dataset is hardly possible for this project, and such dataset collection is not taken up. Live pilot–ATC communication is operational traffic, and it is not available for collection as a dataset in a project of this kind. Therefore, a real-time dataset could not be acquired during this semester, and this is recorded as a limitation of the present work.

7.4.3 Benefits Gained from the Professional Interaction

Although the real-time dataset could not be obtained, the interaction with the professionals was useful to the project:

1. It gave practical insight into how pilot–ATC voice communication is carried out in real operation.
2. It gave a clear idea of the constraints in accessing operational aviation audio.
3. It helped to set a realistic scope for the project and to shift the focus from cockpit speech to pilot–ATC VHF communication.

7.4.4 Impact on the Project and Way Forward

The inability to collect a real-time dataset does not remove the contribution of the framework. The following position is taken for the project:

1. The pilot–ATC VHF communication framework and the proposed receiving system are retained as the design contribution of Project III.
2. The non-availability of a real-time aviation speech dataset is reported openly as a limitation of the work.
3. The Speech Emotion Recognition methodology of Phase I will be applied in later phases, and the acquisition framework can be used if authorised access to aviation speech becomes possible in future.

7.5 SUMMARY OF THE IMPROVEMENTS IN PROJECT III

The improvements made in the project during this semester are summarised below:

1. The limitations of existing SER datasets (RAVDESS, CREMA-D, TESS and SAVEE) for aviation use were identified, which established the need for a domain-specific dataset.
2. The focus was shifted from cockpit speech, which is not accessible, to pilot–ATC VHF radio communication.
3. The aviation voice communication chain (VHF band 118.000 – 136.975 MHz, AM DSB-FC) was studied.
4. A pilot–ATC communication framework with a proposed receiving system for data acquisition was designed (Figure 4).
5. The advice and experience of the airport authority and aviation professionals were sought for real-time data collection; it was found that collecting a real-time dataset is hardly possible, and this has been recorded as a limitation.